

CSA1006-SOFTWARE ENGINEERING

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CO1 AT-1

CASE STUDY

AI-Based Smart Campus Maintenance Management System

A large university has more than 30 academic buildings, laboratories, hostels, libraries, and sports facilities. Students and faculty frequently report maintenance issues such as broken classroom equipment, damaged furniture, water leakage, electrical faults, and internet connectivity problems. Currently, complaints are submitted manually through paper forms or phone calls, causing delays, duplicate requests, and poor communication between departments.

The university plans to develop an **AI-Based Smart Campus Maintenance Management System** that allows users to report issues online, automatically prioritizes complaints using AI, assigns tasks to maintenance staff, tracks repair progress, and provides notifications until the issue is resolved.

1. Understanding the Problem Statement

Problem Explanation

The existing maintenance process is slow, manual, and inefficient. Complaint records are difficult to manage, leading to delayed repairs and reduced user satisfaction.

Objectives

- Develop an online complaint management system.
- Automate complaint prioritization.
- Assign maintenance staff automatically.
- Track repair status in real time.
- Generate maintenance reports.

Scope

- Student complaint registration
- Staff complaint registration
- AI-based priority assignment
- Maintenance tracking
- Admin dashboard
- Notification system
- Report generation

2. Stakeholder Analysis

Stakeholder	Responsibilities
Students	Register maintenance complaints
Faculty	Report classroom/lab issues
Maintenance Staff	Repair assigned complaints
Maintenance Manager	Assign and monitor tasks
Administrator	Manage users and reports
University Management	Monitor overall maintenance performance

Challenges

- Delayed complaint handling
- Duplicate complaints
- Poor communication
- Lack of real-time tracking
- Manual report preparation

3. Current System Analysis

Existing Process

- Complaints submitted manually.
- Staff informed through phone or email.
- Repairs tracked manually.
- Reports prepared using spreadsheets.

Strengths

- Easy to understand
- No special software required

Limitations

- Time-consuming
- Data redundancy
- No automation
- Difficult tracking
- Poor transparency

- **Problems**

- Long repair time
- Complaint duplication
- Lost complaint records
- Inefficient resource allocation

4. Proposed Software Solution

Proposed System

Develop a web-based application with AI-assisted complaint prioritization.

Functional Modules

Module 1: User Management

- Student Login
- Faculty Login
- Admin Login

Module 2: Complaint Registration

- Upload complaint
- Add photo
- Select location
- Submit issue

Module 3: AI Priority Module

- High Priority
- Medium Priority
- Low Priority

Module 4: Maintenance Assignment

- Assign technician
- Update repair status

Module 5: Dashboard

- Complaint statistics
- Pending issues
- Completed repairs

Module 6: Report Generation

- Daily reports
- Monthly reports
- Performance reports

Functional Requirements

- User Registration
- Login
- Complaint Submission
- Complaint Tracking
- Notification System
- Admin Dashboard
- Report Generation

Non-Functional Requirements

- Fast response time
- Secure login
- High reliability
- Easy to use
- Scalable architecture
- Data backup

Software Engineering Principles Used

Modularity

Separate modules for login, complaint, reports, dashboard.

Scalability

Supports increasing users and buildings.

Maintainability

Easy to update modules independently.

Reliability

Automatic backup and validation.

Usability

Simple user interface.

Security

Password encryption and role-based access.

5. SDLC Justification

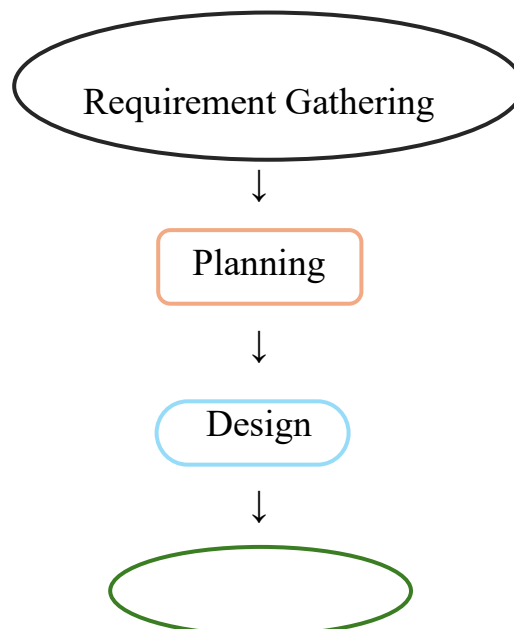
Selected Model

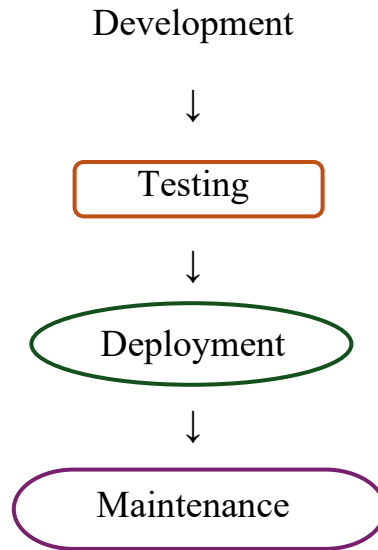
Agile SDLC

Why Agile?

- Requirements may change.
- Easy to add new features.
- Continuous testing.
- Frequent customer feedback.
- Faster delivery.

Agile Process





6. Expected Outcomes

Benefits

- Faster complaint resolution
- Better maintenance planning
- Reduced paperwork
- Improved transparency
- Increased user satisfaction
- Better resource utilization

Addressed Challenges

- Eliminates manual records
- Reduces duplicate complaints
- Provides real-time updates
- Improves communication

Risks

- Internet dependency

- User training required
- Initial implementation cost
- AI prediction errors

Future Enhancements

- Mobile application
- Voice complaint registration
- IoT sensor integration
- Predictive maintenance
- QR-code based complaint registration

Output

Smart Campus Maintenance Management System

Smart Campus Maintenance Management System

Name
Mounika

Email
mounika@simats000

Building
Hostel

Issue Type
Furniture Damage

Description
Fan is not working and water leaking from the Ac

Submit Complaint

Smart Campus Maintenance Management System

Smart Campus Maintenance Management System

Name
Mounika

Email
mounika@simats000

Building
Hostel

Issue Type
Furniture Damage

Description
Fan is not working and water leaking from the Ac

Submit Complaint

Complaint Details

 Complaint Registered Successfully

Complaint ID : CMP2510
Name : Mounika
Email : mounika@simats000
Building : Hostel
Issue : Furniture Damage
Description : Fan is not working and water leaking from the Ac

Priority : Low
Technician : Carpenter
Status : Pending

OK

Conclusion

The proposed system improves the efficiency, transparency, and reliability of campus maintenance management. By using Agile SDLC and software engineering principles, it provides a scalable and user-friendly solution that enhances user satisfaction and can be further improved with AI, mobile apps, and IoT integration in the future.